

MINDTREK

Mindtrek 2026, Tampere, 6th of October, 2026

Sovereign *by habit*

20 years of self-hosting from source
on European servers

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About *Rolle*

Founder and CTO of Digitoimisto Dude Oy. Code soon 30 years, WordPress since 2005, online since 1999. Accessibility, open source, Linux servers, and building my own tools. Free time: running, bilingual family, knows sign language.

DUDE

Contributing to WordPress and open source since we founded Dude in 2013: 71 open source repositories on GitHub, including air-light, our starter theme with over 1100 stars. Self-hosting as much as we can.

Sovereign by habit



What I mean by *server and source*



Server

A computer the outside world can reach online.



Source

A file I can read and study myself, without barriers.

When some of us started, *this was the whole stack*

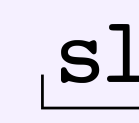
 Commodore

 GNU

 FreeBSD

 Perl

 Linux

 slackware
l i n u x

 debian

 redhat

 openSUSE

 APACHE

 MySQL

 php

 Vim

 Emacs

My first servers *ran at home*

Dynamic DNS

A script updated the DNS the moment my home IP changed.

Renting servers

Anything that needed bandwidth moved out of the wardrobe to rented servers.

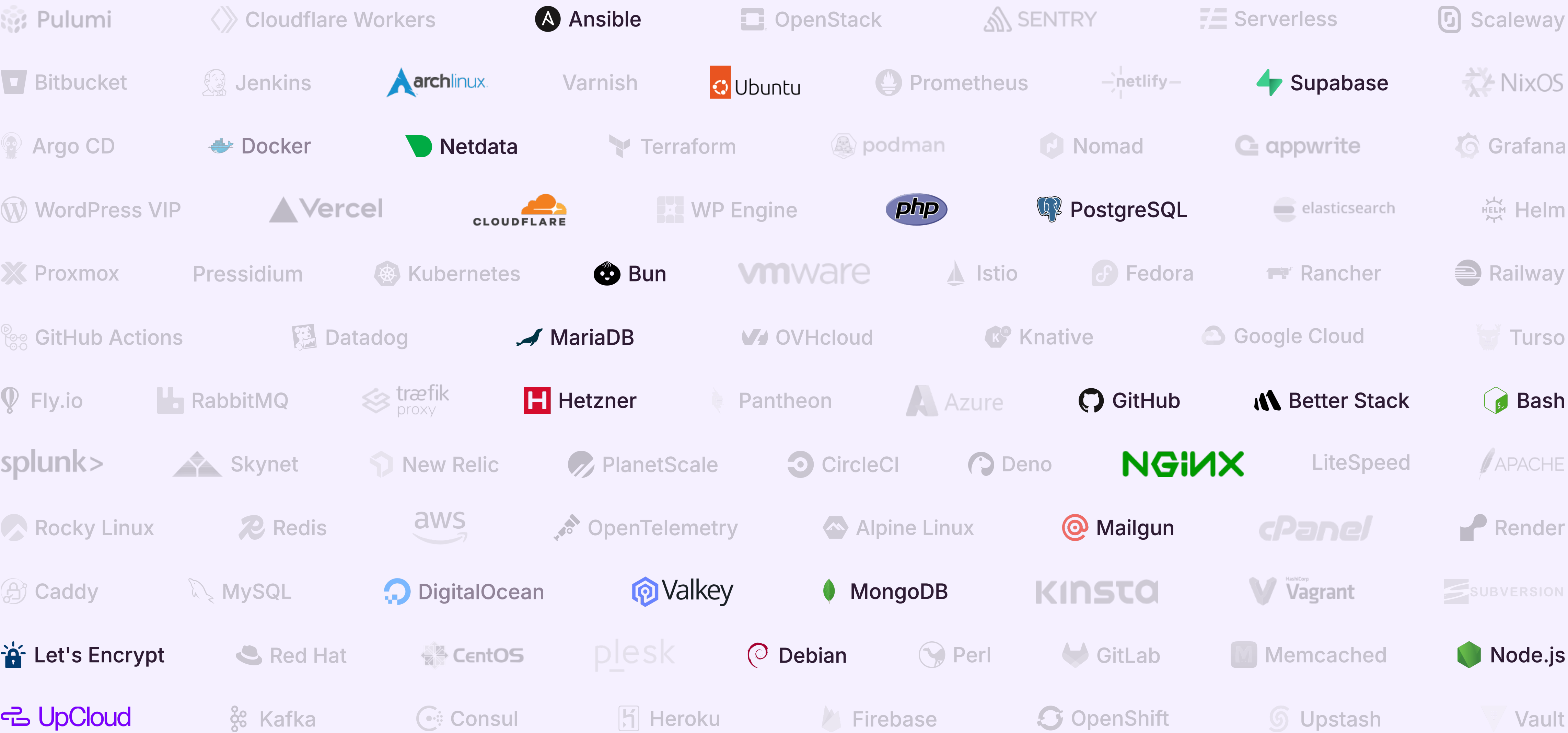
The problems

It ate the family bandwidth, upload was tiny, and it hummed in a wardrobe.

Still at home

My IoT still runs at home on my own hardware, and I still tinker and self-host at home.



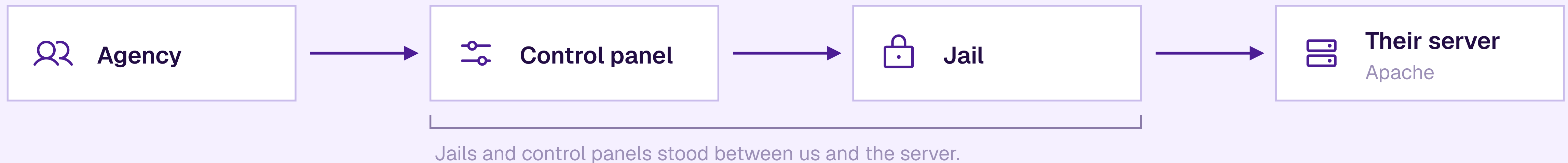


I want to see *the filesystem*

Every system runs inside some box. I want to be able to open mine and read it.

From partners *to our own server*

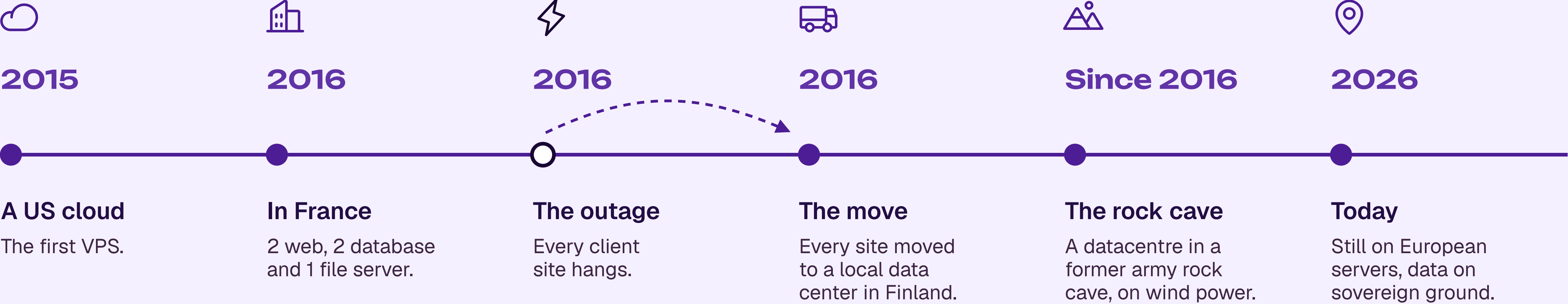
2013 Client sites on hosting partners' servers.



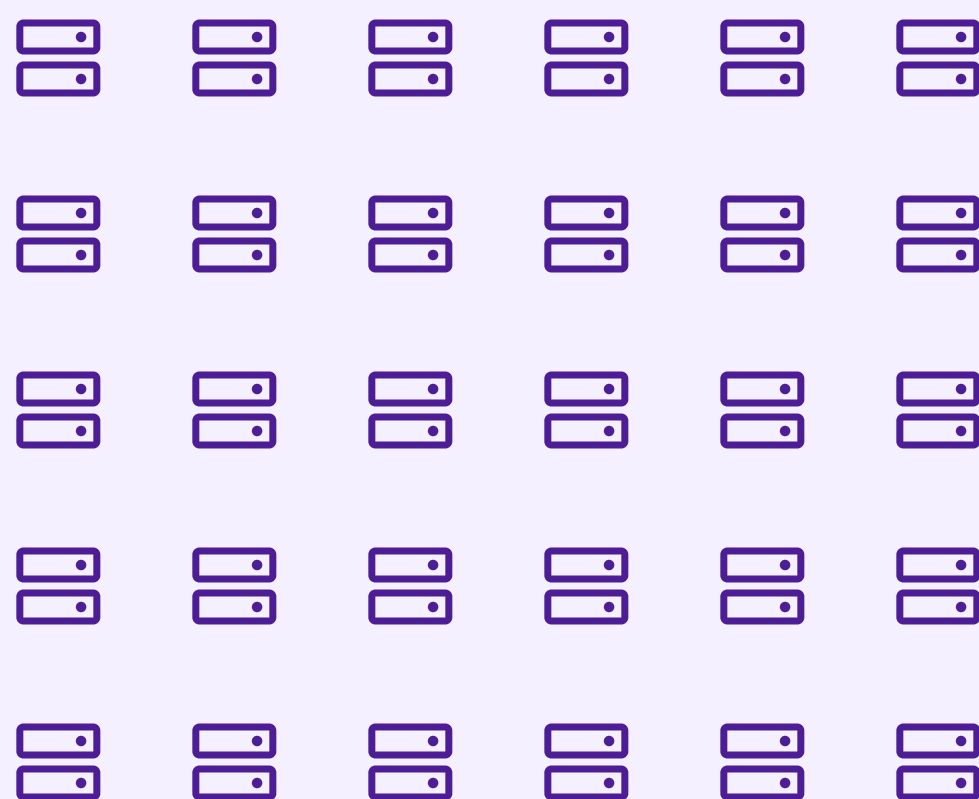
2015 Our first own VPS, to run the latest nginx and HHVM-FastCGI.



Choosing providers, *and leaving them*



The fleet *today*



Dozens of clusters

Self-hosted Linux and virtualization.



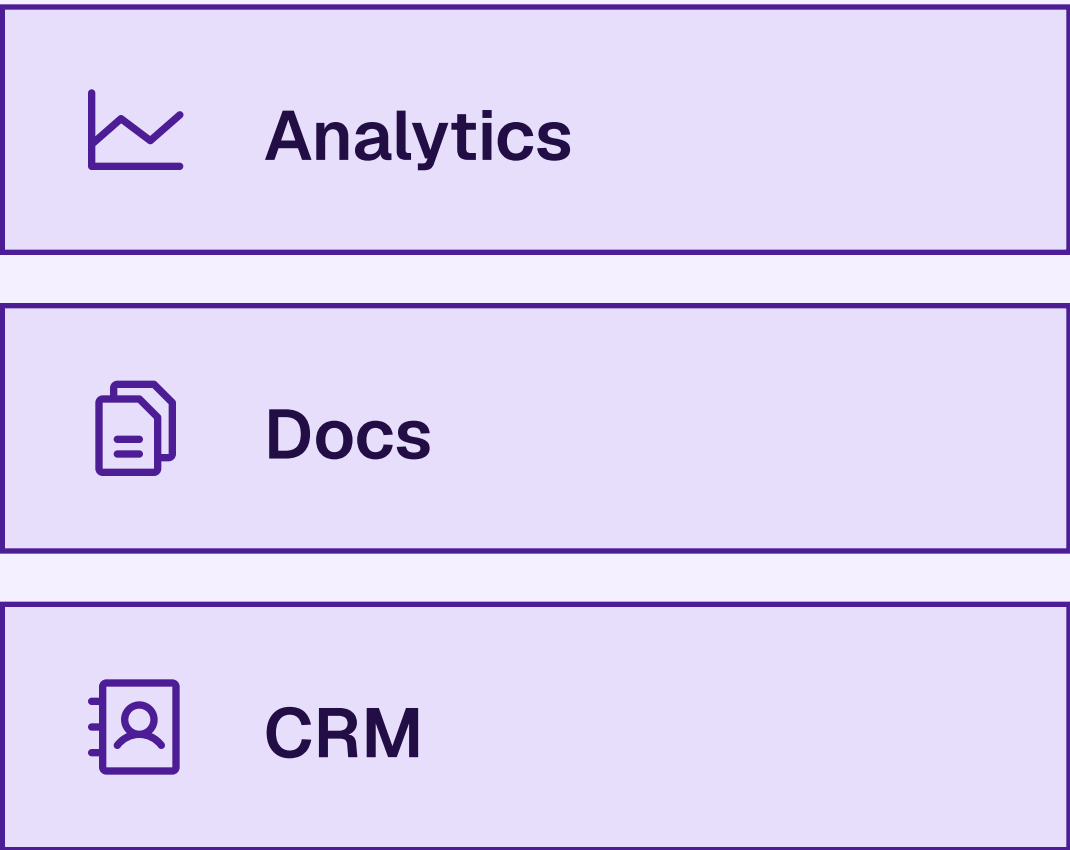
14 people

A web agency with its own sysop team.



Built for purpose

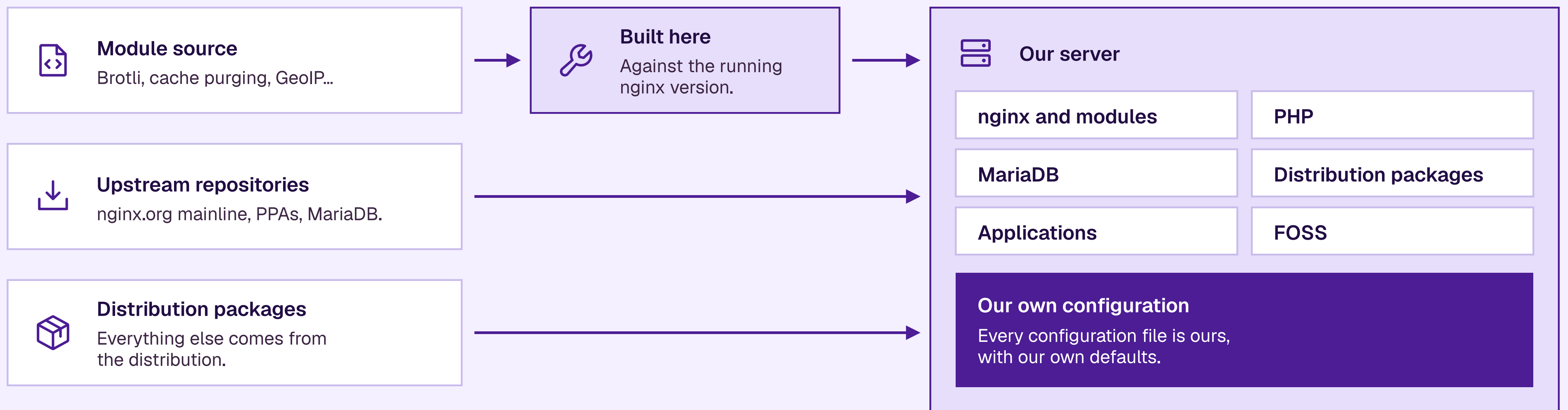
Each server is set up for its use, with nothing extra installed.



Own your services

Analytics, docs, CRM, and other software run on our own servers.

From source *and from upstream*



The package broke, *source kept running*

Read the running build

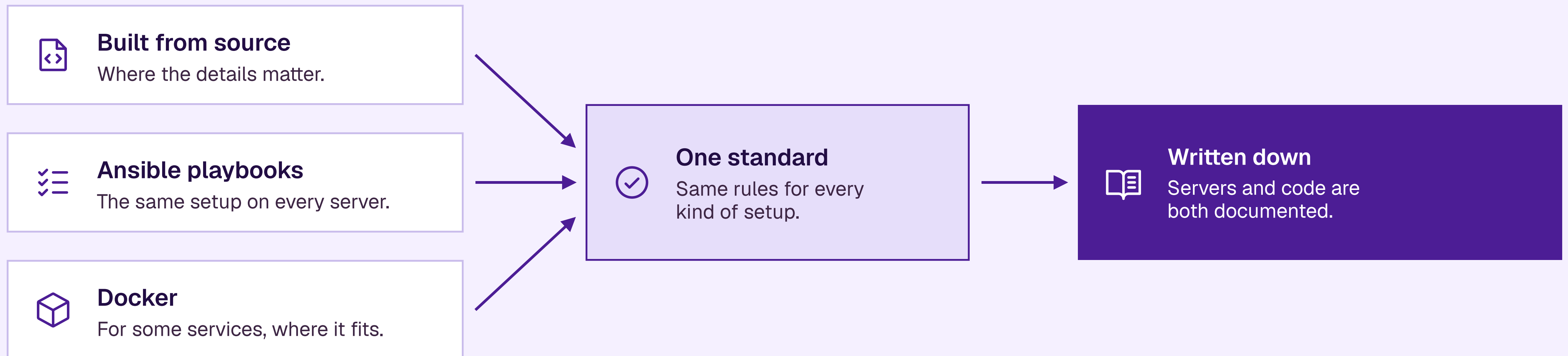
```
nginx -v  
nginx -V 2>&1 | grep configure
```

Build the module against it

```
curl -O nginx.org/download/nginx-$V.tar.gz  
tar xzf nginx-$V.tar.gz && cd nginx-$V  
./configure <same arguments> \  
    --add-dynamic-module=../ngx_cache_purge  
make modules
```

Canonical's nginx security update made nginx crash with external modules. Our nginx comes from nginx.org mainline, with modules built from source against it, and it kept running.

Self-hosting *needs standards too*



Self-hosting *takes more work*

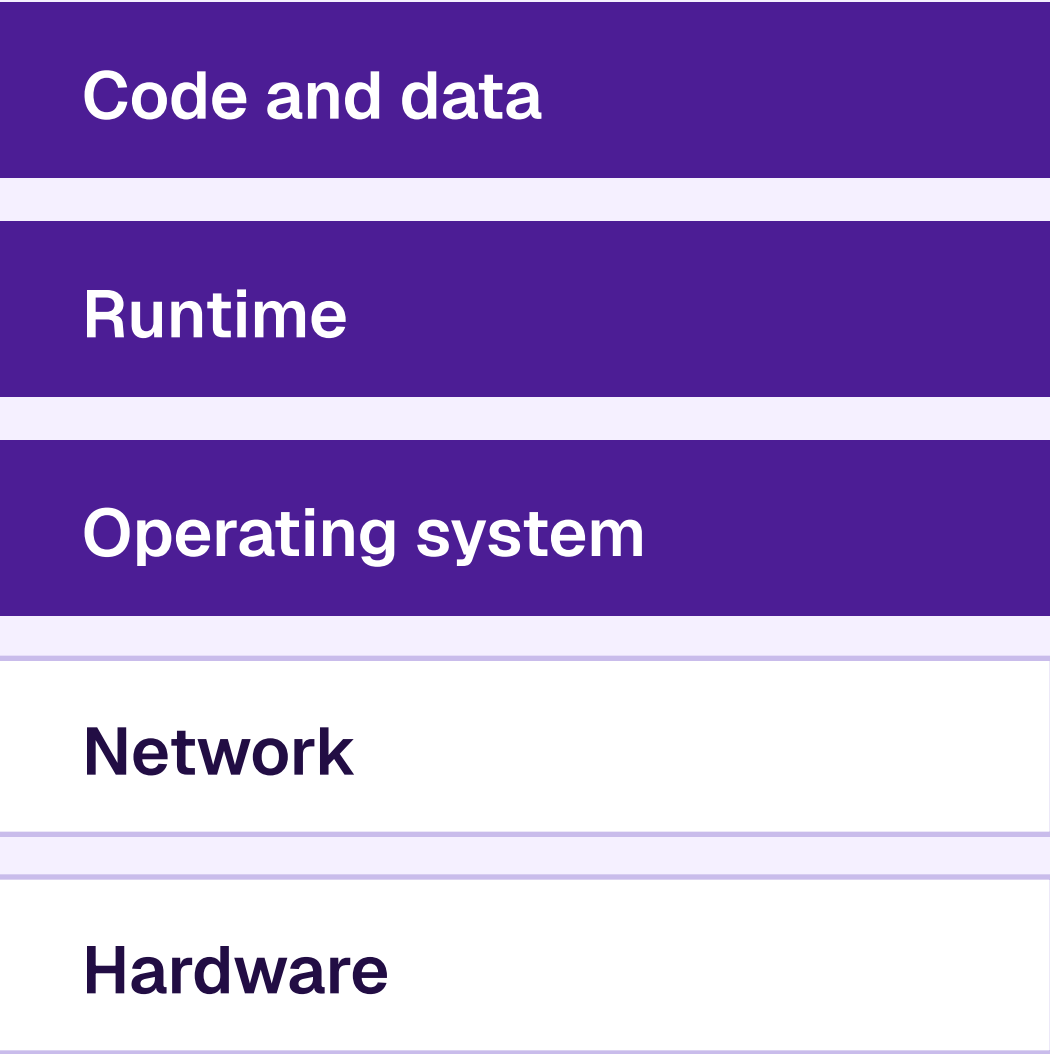
- Most of the work goes into compiling and configuring.
- Servers drift apart without fleet control.
- Fixes happen manually first and end up in scripts later.
- Knowledge stays with a few people unless it is written down.
- AI now speeds up the configuring and the scripting.

Why we *still do it*

- Every layer can be read, down to the source code.
- A crash can be traced from the log to the source.
- Portability and owning your data. In 2016 we moved every site to another provider.
- No vendor lock-in.

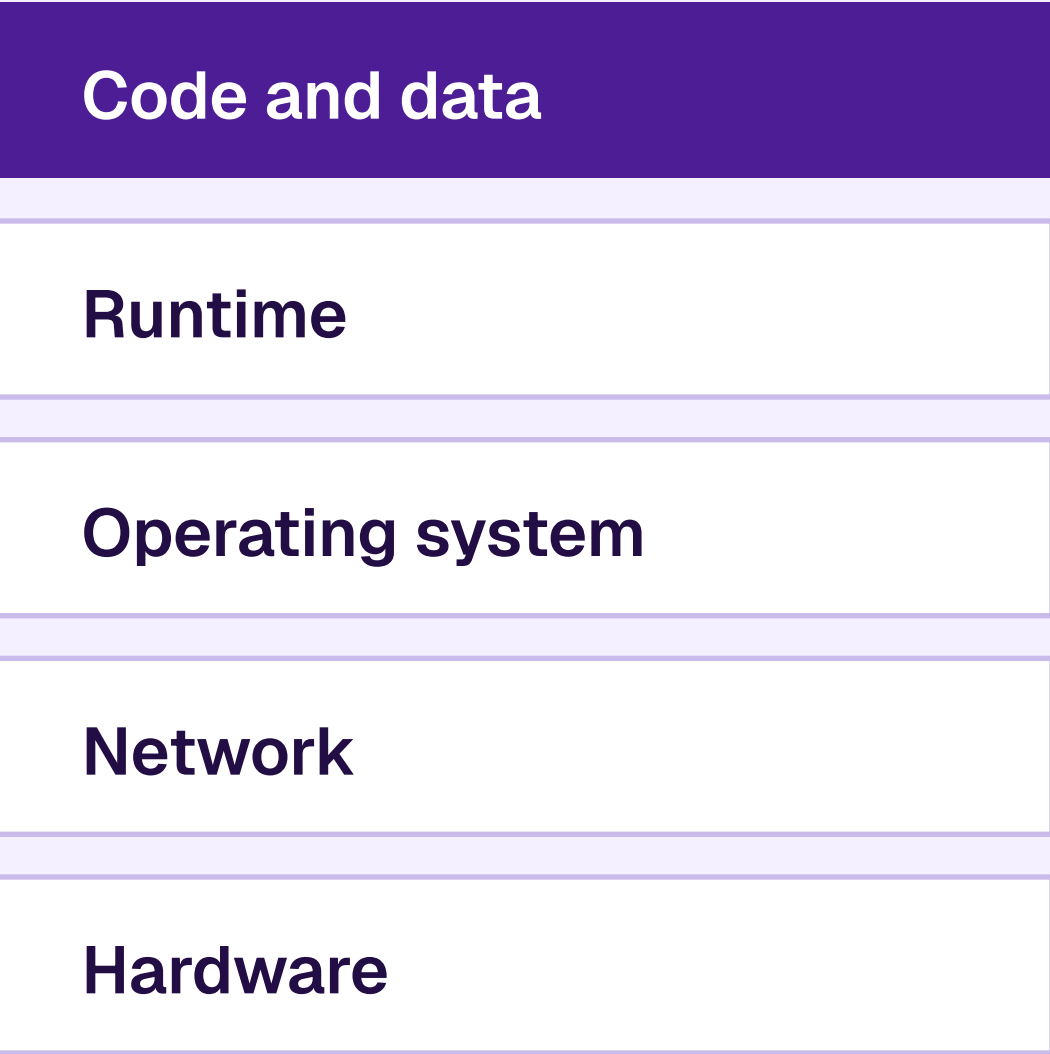
Own servers *and the cloud*

Own servers



- +** We can open every layer, and moving to another provider means copying files and databases.
- We patch, monitor and fix everything ourselves, also at night.

Cloud

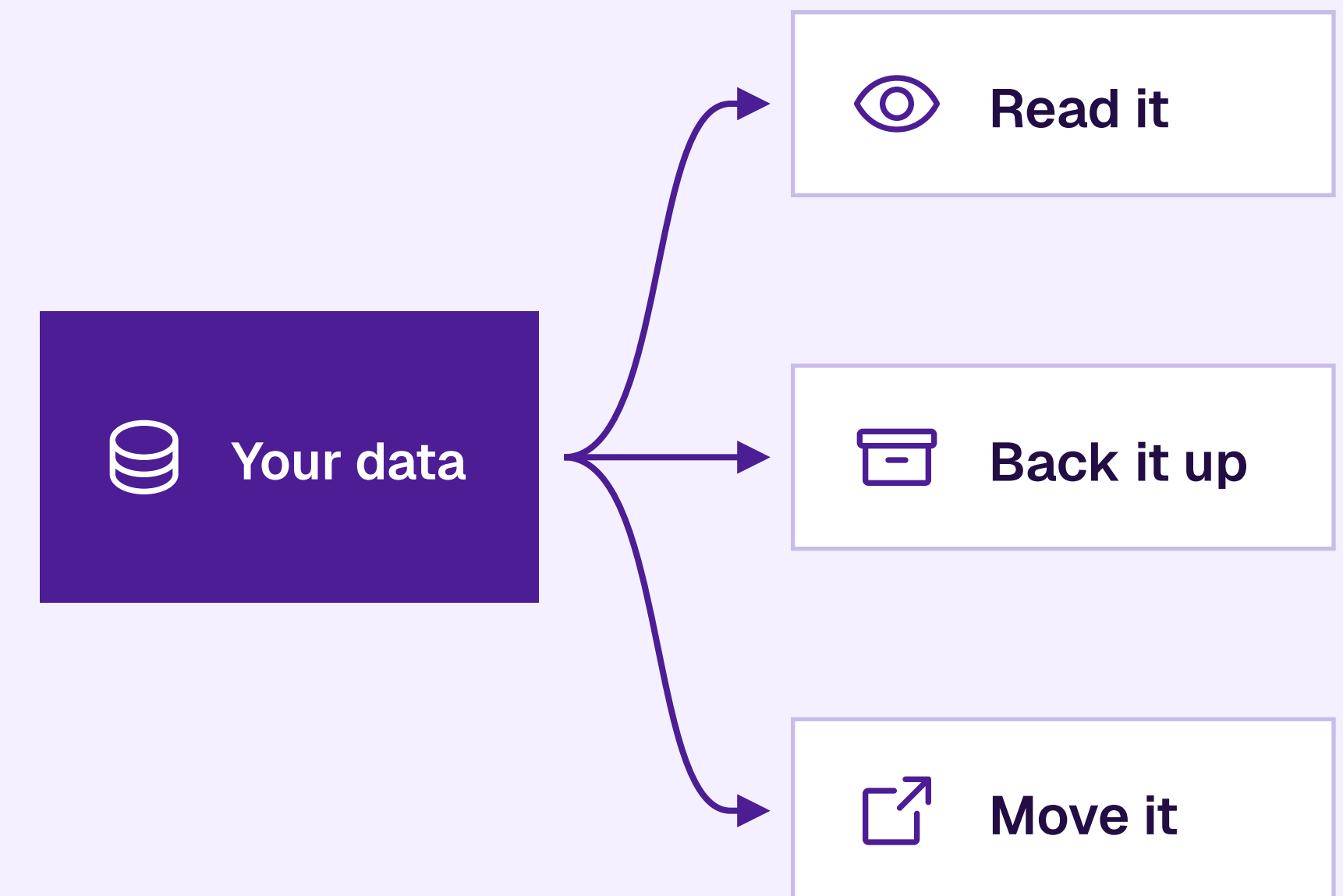


- +** The provider handles hardware failures, failover and scaling.
- When the platform breaks, we wait. An American company's EU region still falls under US law.

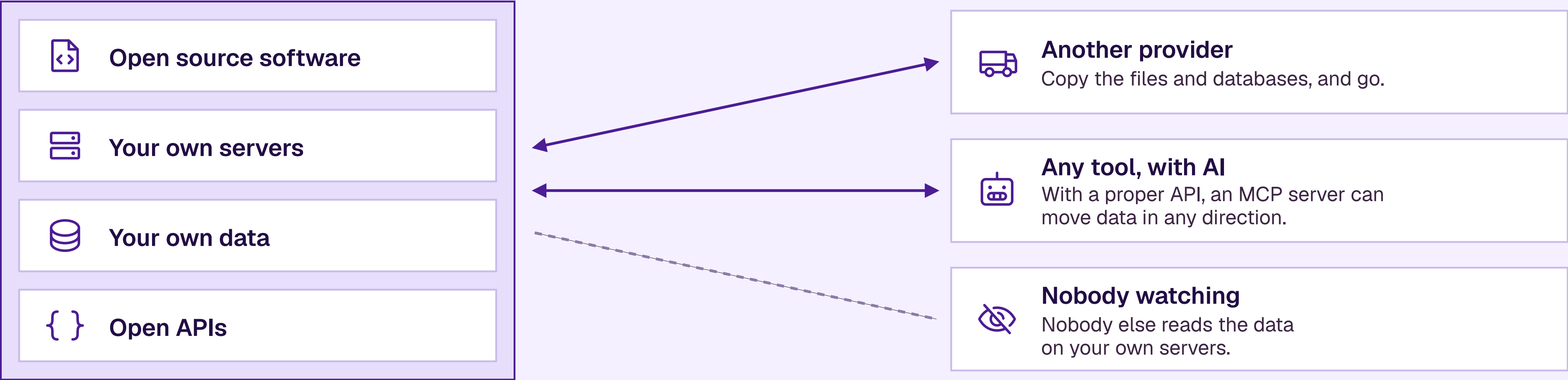
 We are in control  The provider is in control

Own your data *and know where it lives*

FOSS, analytics, documents, CRM all live on servers you own and control, with back ups.



True portability, *move any time*



We still depend *on others*

DNS and CDN

Cloudflare and similar for DNS and caching.

Code and tools

GitHub for code, Better Stack for uptime, Let's Encrypt for certificates.

Email

Mailgun, SendGrid, large services for reputation. Delivery depends on the large mail providers.

AI

Open-source AI requires powerful hardware and costs much more to host.



More at
european-alternatives.eu

Big tech has *alternatives*

Big tech

Cloudflare

Mailgun, SendGrid

GitHub

Google Workspace

OpenAI

European or self-hosted

→ Bunny.net, Slovenia

→ Brevo, France

→ Codeberg, Germany, or Forgejo

→ Nextcloud, self-hosted

→ Mistral, France, or local models

The catch

Smaller network than Cloudflare.

Delivery still depends on Gmail and Outlook.

Most open source projects still live on GitHub.

Someone has to run and update it.

Local models need strong hardware.



Where to *start*

Start here

New to running servers

- 1** Start on your own computer. Use Linux, run one service, and read its config files.
- 2** Rent the smallest VPS (e.g. UpCloud) and set it up yourself: SSH keys, firewall, nginx and TLS.
- 3** Write every step down. Start scripting.

Go further

Already running servers

- 4** Move one service you subscribe to onto your own server: Plausible analytics, Outline docs, etc.
- 5** Swap one installed package for a source build. Try to see if it compiles via a script or Ansible.
- 6** Test your exit. Restore a backup and move one service to another provider.



Knowing every layer *is practical sovereignty*

Host the thing yourself, end to end,
and keep it running for a year.



Kiitos. *Questions?*

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 github.com/rollecode
github.com/rollecode/mindtrek2026-talk-slides

